Typing the Unreliable Participant:
Modes, Taint, and Grades for Goal Achievability of Approximately Compliant Agents

WORKING DRAFT — rules for co-author markup — not for submission

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Abstract

Multiparty session types deliver their guarantees under a hypothesis so standard it is rarely stated: every participant runs code that was checked against its type. A large language model acting as a participant breaks the hypothesis — it *approximately* follows the behaviour it was given, and every classical theorem silently loses its precondition. We extend the goal-marked, capability-guarded session discipline of the skill-achievability compiler with a *deviation layer* that types non-compliance itself. Three empirical phenomena are internalized as structure in the rules, with no probabilities in the type system: correlated errors become per-role *compliance modes* on a three-point lattice, monotonically downgraded by observed deviations and propagated by *taint* through received data; context-dependent degradation becomes a per-role *grade* (steps since last re-grounding) with a checkpoint well-formedness condition on recursion; adversarial deviation (prompt injection) becomes a *demonic bottom mode* with game semantics and a typed irreversibility guard. Numbers enter only through a *compliance profile* — the behavioural sibling of the capability context Γ — interpreted over a Markov-modulated stochastic game, against which the graded achievability judgment $\Diamond_{\geq p}\,\varphi_{goal}$ is sound. Safety is unconditional; achievement is graded. We state the metatheory as seven theorems with explicit proof obligations, a mechanization plan that keeps the qualitative layer in Coq and the game-theoretic layer on paper, and three pre-registered falsifiable predictions that the rule structure makes about measured agent traces.

WIP: This draft fixes the *shape* of the extension: syntax, instrumented reduction rules, typing rules, well-formedness conditions, theorem statements, and the semantic model. Proofs are stated as obligations, not discharged. Open design points are marked WIP inline; the largest is the quarantine rule T-QUAR (Remark 1). Base-rule notation follows `tas.tex` exactly; any conflict there is a bug in this file.

1 Introduction

Classical multiparty session types (MPST) guarantee communication safety, session fidelity, and deadlock-freedom for systems whose participants *are* their local behaviour: the code was typed, the type is the behaviour, and every run is a compliant run. The guarantees are, in other words, universally quantified over compliant runs. An LLM agent occupying a role changes the quantifier: the agent was *instructed* with its behaviour, not compiled from it, and compliance is now a property a run may or may not have. Testing observes runs; typing quantifies over all of them — but over all of *which* runs, when the participant only approximately follows the type?

*Working draft on top of the skill-achievability discipline; base rules follow `paper/tas.tex`, which remains authoritative for Section 2.

The nearest formal answers each miss the phenomenon. Probabilistic session types put probability on the internal *choices* of compliant processes [1, 2, 3, 4]; the dice are rolled exactly where the type permits. Crash-stop and affine disciplines [6, 7] model participants that *stop*; an LLM more often keeps going, wrongly. Runtime monitoring restores safety at the boundary [8] but says nothing about whether the goal of the session is still *achievable* once a participant has begun to drift. And none of these confront the three facts every measured agent trace exhibits:

1. **Correlation.** Errors are not i.i.d. coin flips: one misunderstanding corrupts the remainder of the run. Deviations cluster.

2. **Context-dependence.** Deviation rates grow with the distance from the last re-grounding: depth of recursion, length of accumulated context.

3. **Adversariality.** Prompt injection is not a coin flip at all; an injected participant is best modelled as an adversary, not a noisy channel.

A model that assumes "each step deviates independently with probability ε" is therefore a model of a machine that does not exist, and any theorem proved about it inherits the fiction. Our design principle is to internalize each phenomenon as *structure in the typing rules* while keeping all numbers out of them: probabilities live only in the semantic model, as parameters. The rules carry *modes*, *taint*, and *grades*.

Contributions.

C1 A deviation-instrumented operational semantics for the goal-marked session calculus of the skill-achievability discipline (Section 2 recaps the base system): configurations carry a mode environment, a grade environment, and a taint set; deviant steps are blocked stutters that downgrade modes (Section 3).

C2 A mode-graded type system in which compliance modes descend a three-point lattice by evidence and ascend only by attestation; received data propagates taint; goal markers $\checkmark\varphi$ double as taint sanitizers; a demonic bottom mode types injected participants against arbitrary behaviour; and an irreversibility rule forbids critical effects on tainted data (Section 4).

C3 A checkpoint well-formedness condition on recursion, with a bounded-degradation theorem and its negative converse: uncheckpointed loops admit no nontrivial achievability bound (Section 6).

C4 A semantic model — a Markov-modulated stochastic game parameterized by a *compliance profile* $\mathcal{E}$, the behavioural sibling of Γ — and a graded achievability judgment $\Diamond_{\geq p}\varphi_{goal}$ sound with respect to the game value (Section 5).

C5 A metatheory of seven theorems with explicit proof obligations and a mechanization plan (qualitative layer in Coq, game-theoretic layer on paper), plus three pre-registered falsifiable predictions the rule structure makes about measured agent traces (Sections 6–7).

Conservativity anchors the design (Proposition 1): with all modes `ok`, empty taint, and a zero deviation profile, every rule collapses to its base counterpart and every base theorem is preserved. The deviation layer is a product construction over the base discipline, not a rewrite of it.

2 The Base Discipline in Sixty Seconds

We recap the skill-achievability discipline; `tas.tex` is authoritative for this section. Disjoint sets: predicates $p \in \mathsf{Pred}$, numeric variables $x \in \mathsf{Var}$, roles $\mathsf{p}, \mathsf{q}, \mathsf{r}$, capability names $a \in \mathsf{Cap}$, labels $\ell \in \mathsf{Lab}$. The state logic $\mathcal{L}$ is quantifier-free linear integer arithmetic with uninterpreted boolean predicates. A world $W = \langle B, N \rangle$ valuates predicates and variables. A capability context maps each possessed tool to a guarded STRIPS effect, $\Gamma(a) = \langle \mathit{pre}_a, \mathit{eff}_a \rangle$ with $\mathit{eff}_a = \langle \mathit{Add}_a, \mathit{Del}_a, \mathit{Asg}_a, \mathit{Nd}_a \rangle$; $a \notin \mathrm{dom}(\Gamma)$ means the tool does not exist for this agent.

Processes and global types are coinductive regular trees:

$$P \;::=^{coind}\; \mathbf{0} \;\mid\; \mathsf{p}!\{\ell_i.\, P_i\}_{i\in I} \;\mid\; \mathsf{p}?\{\ell_i.\, P_i\}_{i\in I} \;\mid\; a.\, P$$
$$G \;::=^{coind}\; \mathsf{p}{\to}\mathsf{q} : \{\ell_i.\, G_i\}_{i\in I} \;\mid\; a@\mathsf{p}.\, G \;\mid\; \checkmark\varphi.\, G \;\mid\; \mathsf{End}$$

A session $\mathbb{M} = \mathsf{p}_1[P_1] \parallel \cdots \parallel \mathsf{p}_n[P_n]$ reduces as a configuration $(\mathbb{M}; W)$; only capabilities touch the world (World-Act: $\langle W \rangle\, a\, \langle W' \rangle$ when $W \vDash \mathit{pre}_a$ and $W' \in \mathrm{apply}(\mathit{eff}_a, W)$). The session LTS has two rules:

S-Comm
$$\frac{k \in I \subseteq J}{(\mathsf{p}[\mathsf{q}!\{\ell_i.P_i\}_{i\in I}] \parallel \mathsf{q}[\mathsf{p}?\{\ell_j.Q_j\}_{j\in J}] \parallel \mathbb{M}; W) \xrightarrow{\mathsf{p}\,\ell_k\,\mathsf{q}} (\mathsf{p}[P_k] \parallel \mathsf{q}[Q_k] \parallel \mathbb{M}; W)}$$

S-Act
$$\frac{\langle W \rangle\; a\; \langle W' \rangle}{(\mathsf{p}[a.P] \parallel \mathbb{M}; W) \xrightarrow{a@\mathsf{p}} (\mathsf{p}[P] \parallel \mathbb{M}; W')}$$

Global types have head rules G-Comm-E/G-Act-E/G-Goal-E and interleaving rules G-Comm-I/G-Act-I/G-Goal-F over $(G, W) \rightsquigarrow_\Lambda (G', W')$; achievability is existential may-reachability of a goal-satisfying configuration, $\Gamma; G \vDash \Diamond_{init}\varphi_{goal}$. The whole session is typed *directly* against G — no local types, no projection, no separate subtyping — by the coinductive judgment $G \vdash (\mathbb{M}; W)$ with the four rules T-End, T-Act, T-Goal, T-Comm, each carrying a participant-matching side condition ($\mathsf{prt}(G)$ against $\mathsf{prt}(\mathbb{M})$), receiver slack $I \subseteq J$ inside T-Comm, and plain-merge discipline via the shared-residual premise. The top rule Ach combines no-hallucinated-capabilities, conformance, and reachability. Subject reduction and session fidelity connect the two labelled systems.

Everything above presumes each $\mathsf{p}[P]$ *executes* P. The rest of this paper removes that presumption.

3 Deviation: Extended Packs and Instrumented Semantics

3.1 What a deviation is

The monitored deployment (the gate of the compiler's runtime) observes each role at the message-and-capability boundary; it does not observe reasoning. Relative to a declared behaviour, an agent step can misbehave in exactly these observable ways:

class	attempted move	detected by	effect when gated
δ_{lab}	send a label outside the enabled branch set	gate (EFSM)	blocked stutter
δ_{ord}	send/receive not enabled at the current node	gate (EFSM)	blocked stutter
δ_{cap}	fire a not enabled here, or with $W \nvDash \mathit{pre}_a$	gate (Γ, W)	blocked stutter
δ_{pay}	effect outside its declared Nd envelope	only later, by $\checkmark\varphi$	*latent*: taint
δ_{sil}	no move (abandonment/timeout)	scheduler	stutter, downgrade

The first three and the last are caught at the boundary and become *blocked stutters*; only δ_{pay} is latent, which is exactly why taint exists: data whose validity the gate could not decide at write time is remembered as suspect until a $\checkmark\varphi$ check discharges it.

3.2 Extended packs

A *deviation-aware pack* extends the base pack $\langle \Gamma, G, \varphi_{goal}, W_0 \rangle$ with declarations that are all checkable at the schema gate:

$$\mathcal{P} \;=\; \langle\, \Gamma,\; G,\; \varphi_{goal},\; W_0,\; \mathsf{Crit},\; \mathsf{prov},\; \mathsf{Att},\; \mathsf{Rfr} \,\rangle$$

- $\mathsf{Crit} \subseteq \mathsf{Pred}$: *critical atoms*. A capability is *irreversible*, $a \in \mathsf{Irr}$, iff $Del_a \cap \mathsf{Crit} \neq \emptyset$ or a carries a declared external-write flag.
- $\mathsf{prov} : \mathrm{dom}(\Gamma) \to \{\mathsf{int}, \mathsf{ext}\}$: *provenance*. An ext capability's writes carry data originating outside the session (web fetch, file read, tool output).
- $\mathsf{Att} \subseteq \mathrm{dom}(\Gamma)$: *attestation capabilities* (human sign-off, test execution). Well-formedness WF-Att: $\mathsf{Att} \cap \mathsf{Irr} = \emptyset$ and $\mathsf{prov}(a) = \mathsf{int}$ for $a \in \mathsf{Att}$. An attestation names the role it vouches for; we write $a^{\uparrow \mathsf{q}}$.
- $\mathsf{Rfr} \subseteq \mathrm{dom}(\Gamma)$: *refresh capabilities* (re-read the spec, compact the context). They have empty world effect (eff_a is the identity) and exist to reset grades.

3.3 Modes, grades, taint

Definition 1 (Compliance lattice). $\mathbb{M}_{\mathsf{c}} = \{\mathsf{ok} \sqsupset \mathsf{dr} \sqsupset \bot\}$: *aligned*, *drifted*, *compromised*. Downgrade $\downarrow(\mathsf{ok}) = \mathsf{dr}$, $\downarrow(\mathsf{dr}) = \bot$, $\downarrow(\bot) = \bot$; attested raise $\uparrow(\bot) = \mathsf{dr}$, $\uparrow(\mathsf{dr}) = \mathsf{ok}$, $\uparrow(\mathsf{ok}) = \mathsf{ok}$. Meet $\sqcap$ is the lattice meet.

Definition 2 (Extended configuration). $K = \langle \mathbb{M};\, W;\, M;\, d;\, T \rangle$ where $M : \mathsf{prt}(G) \to \mathbb{M}_{\mathsf{c}}$ is the *mode environment* (distinct from the session $\mathbb{M}$), $d : \mathsf{prt}(G) \to \{0, \ldots, D, \infty\}$ the *grade environment* (steps since last re-grounding, saturating at the cap D — the same one-point widening move the base checker uses for numerics), and $T \subseteq \mathsf{Pred} \cup \mathsf{Var}$ the *taint set* (state written under suspicion). Initially $M_0 \equiv \mathsf{ok}$, $d_0 \equiv 0$, $T_0 = \emptyset$.

Grade ticks are written $d' = d{+_{\mathsf{p}}}1$ (increment $d(\mathsf{p})$, saturating). $\mathrm{wr}(a) = Add_a \cup Del_a \cup \mathrm{dom}(Asg_a) \cup \mathrm{dom}(Nd_a)$ is the write set of a capability; $\mathrm{supp}(\varphi)$ the atoms and variables a formula mentions.

3.4 Instrumented reduction rules

The instrumented LTS refines the base LTS: every base transition has a *compliant* instrumented form that additionally updates M, d, T, and new *deviant* transitions represent the gate-visible misbehaviours of Section 3.1. Contamination is a semantic bookkeeping rule of the monitor, not a typing fiction: the monitor *knows* the sender's mode when it delivers a message.

I-Comm

$$\frac{k \in I \subseteq J \qquad M(\mathsf{p}) \neq \bot \qquad M(\mathsf{q}) \neq \bot \qquad M' = M[\mathsf{q} \mapsto M(\mathsf{q}) \sqcap M(\mathsf{p})] \text{ if } M(\mathsf{p}) \sqsubseteq \mathsf{dr}, \text{ else } M}{\langle \mathsf{p}[\mathsf{q}!\{\ell_i.P_i\}_{i\in I}] \parallel \mathsf{q}[\mathsf{p}?\{\ell_j.Q_j\}_{j\in J}] \parallel \mathbb{M};\, W;\, M;\, d;\, T\rangle \xrightarrow{\mathsf{p}\,\ell_k\,\mathsf{q}} \langle \mathsf{p}[P_k] \parallel \mathsf{q}[Q_k] \parallel \mathbb{M};\, W;\, M';\, d{+_\mathsf{p}}1{+_\mathsf{q}}1;\, T\rangle}$$

I-Act

$$\frac{\begin{array}{c}\langle W\rangle\; a\; \langle W'\rangle \qquad M(\mathsf{p}) \neq \bot \qquad a \in \mathsf{Irr} \implies \mathrm{supp}(pre_a) \cap T = \emptyset \wedge M(\mathsf{p}) \sqsupseteq \mathsf{dr} \\ T' = T \cup \mathrm{wr}(a) \text{ if } M(\mathsf{p}) \sqsubseteq \mathsf{dr} \vee \mathsf{prov}(a) = \mathsf{ext}, \text{ else } T\end{array}}{\langle \mathsf{p}[a.P] \parallel \mathbb{M};\, W;\, M;\, d;\, T\rangle \xrightarrow{a@\mathsf{p}} \langle \mathsf{p}[P] \parallel \mathbb{M};\, W';\, M;\, d{+_\mathsf{p}}1;\, T'\rangle}$$

I-Refresh

$$\frac{a \in \mathsf{Rfr} \qquad \langle W\rangle\; a\; \langle W\rangle}{\langle \mathsf{p}[a.P] \parallel \mathbb{M};\, W;\, M;\, d;\, T\rangle \xrightarrow{a@\mathsf{p}} \langle \mathsf{p}[P] \parallel \mathbb{M};\, W;\, M;\, d[\mathsf{p} \mapsto 0];\, T\rangle}$$

I-Att

$$\frac{a \in \mathsf{Att} \qquad M(\mathsf{p}) = \mathsf{ok} \qquad \langle W\rangle\; a\; \langle W'\rangle \qquad \mathrm{supp}(pre_a) \cap T = \emptyset}{\langle \mathsf{p}[a^{\uparrow\mathsf{q}}.P] \parallel \mathbb{M};\, W;\, M;\, d;\, T\rangle \xrightarrow{a@\mathsf{p}} \langle \mathsf{p}[P] \parallel \mathbb{M};\, W';\, M[\mathsf{q} \mapsto \uparrow(M(\mathsf{q}))];\, d{+_\mathsf{p}}1;\, T\rangle}$$

I-Dev

$$\frac{\alpha \text{ attempted by } \mathsf{p} \text{ not enabled in } K \quad (\text{classes } \delta_{\mathsf{lab}}, \delta_{\mathsf{ord}}, \delta_{\mathsf{cap}}, \delta_{\mathsf{sil}})}{K = \langle \mathbb{M};\, W;\, M;\, d;\, T\rangle \xrightarrow{\mathsf{dev}(\mathsf{p})} \langle \mathbb{M};\, W;\, M[\mathsf{p} \mapsto \downarrow(M(\mathsf{p}))];\, d{+_\mathsf{p}}1;\, T\rangle}$$

I-Quar

$$\frac{M(\mathsf{p}) = \bot \qquad \alpha \text{ attempted by } \mathsf{p}}{K \xrightarrow{\mathsf{dev}(\mathsf{p})} K \quad (\text{every move of a } \bot\text{-role is refused; only its attestation/escalation path remains})}$$

Three points deserve emphasis. First, I-Dev is a *stutter*: the blocked message "is never delivered, never advances anyone's state, and never becomes context for another agent" — the session and world are untouched; only the monitor's own state (mode, grade) records the event. Second, the irreversibility premise of I-Act is the operational form of authorization-before-irreversible: an Irr capability cannot fire on tainted support nor from a role below dr, *no matter what the agent attempts*. Third, δ_{pay} has no rule of its own: a payload deviation *is* an ordinary I-Act whose *Nd* outcome the gate cannot audit — that is precisely why its writes enter T when the writer is degraded or the source external, and why only $\checkmark\varphi$ can remove them (Section 4).

On the type side, deviations are invisible by design: the global type does not change under a stutter. The extended global configuration is (G, W, M, d, T); each base rule G-Comm-E/G-Act-E/G-Goal-E and each interleaving rule lifts pointwise with the same $M/d/T$ updates as its I-counterpart, plus one new rule mirroring I-Dev:

G-Dev

$$\frac{\mathsf{p} \in \mathsf{prt}(G)}{(G, W, M, d, T) \rightsquigarrow_{\mathsf{dev}(\mathsf{p})} (G, W, M[\mathsf{p} \mapsto \downarrow(M(\mathsf{p}))], d{+_\mathsf{p}}1, T)}$$

G-Goal-San

$$\frac{W \vDash \varphi \qquad \mathrm{supp}(\varphi) \text{ decided in } W}{(\checkmark\varphi.\, G, W, M, d, T) \rightsquigarrow_{\checkmark\varphi} (G, W, M, d, T \setminus \mathrm{supp}(\varphi))}$$

G-Goal-San refines G-Goal-E: discharging a goal marker whose formula the trusted monitor evaluates against the world *sanitizes* the atoms it checked. The marker keeps its base role in achievability unchanged; sanitization is the second job.

4 The Mode-Graded Type System

The judgment extends direct conformance with the three environments:

$$G \vdash \langle \mathbb{M};\, W;\, M;\, d;\, T\rangle$$

read: "G types session $\mathbb{M}$ from world W under modes M, grades d, taint T." It is coinductive, like the base judgment; the mode and grade components range over finite sets, so the extension preserves regularity (finitely many distinct judgment "states" per node of G).

4.1 Communication under downgrade and contamination

$$\textsc{T-Comm-M}\ \dfrac{\begin{array}{c}\forall i \in I \subseteq J.\ \ G_i \vdash \langle \mathsf{p}[P_i] \parallel \mathsf{q}[Q_i] \parallel \mathbb{M};\, W;\, M \rhd (\mathsf{p},\mathsf{q});\, d{+_{\mathsf{p}}}1{+_{\mathsf{q}}}1;\, T\rangle \\ G^\circ \vdash \langle \mathbb{M}^\circ;\, W;\, M[\mathsf{p} \mapsto {\downarrow}(M(\mathsf{p}))];\, d{+_{\mathsf{p}}}1;\, T\rangle \\ M(\mathsf{p}) \neq \bot \quad M(\mathsf{q}) \neq \bot \quad \mathsf{prt}(\mathsf{p}{\to}\mathsf{q} : \{\ell_i.\, G_i\}_{i\in I}) = \mathsf{prt}(\mathbb{M}) \cup \{\mathsf{p},\mathsf{q}\}\end{array}}{\mathsf{p}{\to}\mathsf{q} : \{\ell_i.\, G_i\}_{i\in I} \vdash \langle \mathsf{p}[\mathsf{q}!\{\ell_i.P_i\}_{i\in I}] \parallel \mathsf{q}[\mathsf{p}?\{\ell_j.Q_j\}_{j\in J}] \parallel \mathbb{M};\, W;\, M;\, d;\, T\rangle}$$

where the *contamination update* is

$$M \rhd (\mathsf{p},\mathsf{q}) \;=\; \begin{cases} M[\mathsf{q} \mapsto M(\mathsf{q}) \sqcap M(\mathsf{p})] & \text{if } M(\mathsf{p}) \sqsubseteq \mathsf{dr} \\ M & \text{otherwise} \end{cases}$$

and $(G^\circ, \mathbb{M}^\circ)$ abbreviates *the same node*: the second premise is the *downgrade-closure* premise — the configuration must remain typable at this node after a blocked deviation by the sender (I-Dev stutters, so type and session are unchanged; only M drops). Because $\downarrow$ strictly descends a three-point lattice, the closure premise unfolds at most twice per role before reaching $\bot$, where T-Quar takes over; typability remains decidable on the regular tree.

The base rule's content is intact: sender offers exactly I, receiver may accept more ($I \subseteq J$), all branches check against the same bystanders $\mathbb{M}$ (plain merge), and the participant-matching condition is verbatim. What is new is that *who you last listened to* is part of the typing state: receiving from a drifted sender drags the receiver to dr — unless the protocol validates first, which is the whole point of placing $\checkmark\varphi$ nodes after untrusted exchanges.

4.2 Actions, grades, refresh

$$\textsc{T-Act-M}\ \dfrac{\begin{array}{c}\exists W'.\ \langle W\rangle\, a\, \langle W'\rangle \qquad \forall W'.\ \langle W\rangle\, a\, \langle W'\rangle \implies G \vdash \langle \mathsf{p}[P] \parallel \mathbb{M};\, W';\, M;\, d{+_{\mathsf{p}}}1;\, T \oplus_a M\rangle \\ G^\circ \vdash \langle \mathbb{M}^\circ;\, W;\, M[\mathsf{p} \mapsto {\downarrow}(M(\mathsf{p}))];\, d{+_{\mathsf{p}}}1;\, T\rangle \\ M(\mathsf{p}) \neq \bot \quad a \in \mathsf{Irr} \implies \mathrm{supp}(pre_a) \cap T = \emptyset \wedge M(\mathsf{p}) \sqsupseteq \mathsf{dr} \quad \mathsf{prt}(G) \cup \{\mathsf{p}\} = \mathsf{prt}(\mathbb{M}) \cup \{\mathsf{p}\}\end{array}}{a@\mathsf{p}.\, G \vdash \langle \mathsf{p}[a.P] \parallel \mathbb{M};\, W;\, M;\, d;\, T\rangle}$$

with the *taint update*

$$T \oplus_a M \;=\; \begin{cases} T \cup \mathrm{wr}(a) & \text{if } M(\mathsf{p}) \sqsubseteq \mathsf{dr} \ \vee\ \mathsf{prov}(a) = \mathsf{ext} \\ T & \text{otherwise.} \end{cases}$$

The base rule's existential/universal pairing over Nd successors is verbatim. The side condition marked $a \in \mathsf{Irr}$ is T-Irreversible as a premise: an irreversible capability types only on untainted support and only from a role at dr or better. Note what this rules out statically: a protocol in which web-fetched data can reach the guard of a destructive tool without passing a $\checkmark\varphi$ node has *no* *typing derivation* — the compiler refutes it before any agent runs, with the refutation naming the untainted-support obligation that fails.

$$\textsc{T-Refresh}\ \dfrac{a \in \mathsf{Rfr} \qquad G \vdash \langle \mathsf{p}[P] \parallel \mathbb{M};\ W;\ M;\ d[\mathsf{p} \mapsto 0];\ T\rangle \qquad \mathsf{prt}(G) \cup \{\mathsf{p}\} = \mathsf{prt}(\mathbb{M}) \cup \{\mathsf{p}\}}{a@\mathsf{p}.\, G \vdash \langle \mathsf{p}[a.P] \parallel \mathbb{M};\ W;\ M;\ d;\ T\rangle}$$

$$\textsc{T-Att}\ \dfrac{\begin{array}{c} a \in \mathsf{Att} \qquad M(\mathsf{p}) = \mathsf{ok} \qquad \mathrm{supp}(pre_a) \cap T = \emptyset \\ \forall W'.\ \langle W\rangle\, a\, \langle W'\rangle \implies G \vdash \langle \mathsf{p}[P] \parallel \mathbb{M};\ W';\ M[\mathsf{q} \mapsto \uparrow(M(\mathsf{q}))];\ d{+_{\mathsf{p}}}1;\ T\rangle \\ \mathsf{prt}(G) \cup \{\mathsf{p}\} = \mathsf{prt}(\mathbb{M}) \cup \{\mathsf{p}\} \end{array}}{a^{\uparrow\mathsf{q}}@\mathsf{p}.\, G \vdash \langle \mathsf{p}[a.P] \parallel \mathbb{M};\ W;\ M;\ d;\ T\rangle}$$

Modes go down by evidence (T-Comm-M's closure premise, G-Dev); they go up *only* through T-Att, whose own preconditions are strict: the attester must itself be ok, and the attestation evidence must be untainted. That asymmetry — cheap descent, expensive ascent — is the design. Data validation and mode restoration are deliberately different rules: $\checkmark\varphi$ cleans *data* (below); only attestation restores *trust in a role*.

4.3 Goal markers as sanitizers

$$\textsc{T-Goal-San}\ \dfrac{\begin{array}{c} G \vdash \langle \mathbb{M};\ W;\ M;\ d;\ T \setminus \mathrm{supp}(\varphi)\rangle \\ W \vDash \varphi \qquad \mathrm{supp}(\varphi) \text{ decided in } W \qquad \mathsf{prt}(G) = \mathsf{prt}(\mathbb{M}) \end{array}}{\checkmark\varphi.\, G \vdash \langle \mathbb{M};\ W;\ M;\ d;\ T\rangle}$$

This is base T-Goal plus the taint discharge: a check the trusted monitor can actually decide against the world clears the atoms it inspected. $\checkmark\varphi$ thus has two jobs — goal credit (unchanged) and sanitization — and the protocol author controls *where* trust is re-established by where markers are placed.

4.4 The demonic bottom

$$\textsc{T-Quar}\ \dfrac{\begin{array}{c} M(\mathsf{p}) = \bot \qquad G \downharpoonleft_{\mathsf{p}} \vdash\ \langle \mathbb{M};\ W;\ M;\ d;\ T\rangle \\ \forall a \in \mathsf{Irr} \text{ with } a@\mathsf{p} \in \mathsf{cap}(G):\ \text{unreachable in } G \downharpoonleft_{\mathsf{p}}\ \text{before} \uparrow \end{array}}{G \vdash \langle \mathsf{p}[P] \parallel \mathbb{M};\ W;\ M;\ d;\ T\rangle}$$

Remark 1 (Open design point: the quarantine residual). WIP. $G \downharpoonleft_{\mathsf{p}}$ is the *quarantine residual* of G: the protocol with every interaction of p replaced by its declared escalation branch (crash-handling style), pending either an attestation $a^{\uparrow\mathsf{p}}$ or End. Two candidate formulations exist: (a) require every choice node involving p to carry an explicit escalation branch (syntactic; heavier protocols; trivially sound), or (b) define $\downharpoonleft$ as a partial function whose undefinedness refutes achievability with a new reason code (Unquarantinable_Role), in the spirit of undefined merge. We lean (b) — it matches the discipline's refutation asymmetry — but the co-author should rule. The demonic quantification itself needs no rule beyond I-Quar: at $\bot$ every attempted move is refused, so "typed against arbitrary behaviour" means the residual system must be typable with p contributing nothing but its escalation path.

4.5 Well-formedness

Definition 3 (Checkpointed recursion)**.** WF-Loop: a global type G (a regular tree) is *checkpointed* iff every cycle of G contains at least one $\checkmark\varphi$ node or one $a@\mathsf{p}$ with $a \in \mathsf{Rfr}$. A per-role refinement (every role active in the cycle is refreshed or validated in it) is strictly stronger; we state theorems for the basic condition and note where the refinement tightens constants.

Definition 4 (Attestation sanity)**.** WF-ATT: $\mathsf{Att} \cap \mathsf{Irr} = \emptyset$; $\mathsf{prov}(a) = \mathsf{int}$ for all $a \in \mathsf{Att}$; every $a^{\uparrow \mathsf{q}}$ occurrence in G is preceded on every path by a $\checkmark\varphi$ whose support covers the evidence atoms of pre_a.

Both conditions are decidable on the regular tree by the same cycle analysis the base checker already performs for loop saturation.

4.6 The graded achievability judgment

Safety is unconditional; achievement is graded. The top rule refines ACH: conformance is checked once, qualitatively; the quantitative bound comes from the semantic model of Section 5.

$$\textsc{Ach-M}\ \frac{\begin{array}{c}\mathsf{caps}(G) \subseteq \mathrm{dom}(\Gamma) \\ \exists W_0 \vDash init.\ \ G \vdash \langle \mathsf{p}_1[S_{\mathsf{p}_1}] \parallel \cdots \parallel \mathsf{p}_n[S_{\mathsf{p}_n}];\, W_0;\, M_0;\, d_0;\, \emptyset\rangle \ \wedge\ \mathrm{val}_{\mathcal{E}}(G, W_0)\ \geq\ p\end{array}}{\Gamma; \mathcal{E} \vdash \{S_{\mathsf{p}}\}_{\mathsf{p}}\ :\ G\ \triangleright\ \Diamond_{\geq p}\, \varphi_{goal}}$$

The base judgment is the p-erasure: dropping the value premise and the three environments recovers ACH exactly. The verdict vocabulary refines accordingly: IMPOSSIBLE (no derivation; a proof, as before), ACHIEVABLE$_{\geq p}$ (derivation plus bound), UNKNOWN (abstention, as before).

5 The Semantic Model: a Markov-Modulated Stochastic Game

Numbers enter here and only here.

Definition 5 (Compliance profile)**.** A *compliance profile* is $\mathcal{E} = \langle \varepsilon,\ \kappa,\ D\rangle$ where $\varepsilon : (\mathbb{M}_{\mathsf{c}} \setminus \{\bot\}) \times \{0, \ldots, D\} \to [0, 1]$ gives the per-step deviation probability of a role in mode m at grade d, monotone in d and antitone in m ($\varepsilon(\mathsf{ok}, d) \leq \varepsilon(\mathsf{dr}, d)$ pointwise is *not* required — drifted roles deviate more: $\varepsilon(\mathsf{ok}, d) \leq \varepsilon(\mathsf{dr}, d)$); κ is a kernel distributing a deviation over the classes of Section 3.1; and D the grade cap. $\bot$ has no ε: compromised roles are not stochastic.

$\mathcal{E}$ is the behavioural sibling of Γ: the capability context says what an agent *can* do, the compliance profile what it *will* do. Both are pack inputs; neither appears in any typing rule. Profiles are *calibrated from measured traces* per model and task family (Section 7), and a profile may be an interval (upper/lower ε), in which case all bounds below hold for the pessimistic end — the imprecise-probability reading.

Definition 6 (The game $\mathcal{G}(\mathcal{P}, \mathcal{E})$)**.** States are extended global configurations (G, W, M, d, T) plus a scheduling component. Three players move:

- the *orchestrator* (angelic) resolves scheduling and internal choice among enabled compliant moves — the existential $\Diamond$ of the base discipline;
- *nature* (probabilistic): a scheduled role p with $M(\mathsf{p}) \neq \bot$ follows the protocol with probability $1 - \varepsilon(M(\mathsf{p}), d(\mathsf{p}))$ and otherwise emits a deviation drawn from κ, which steps by I-DEV (blocked stutter, downgrade) or, for δ_{pay}, perturbs an Nd outcome within the gate-invisible envelope;
- the *adversary* (demonic) controls every role at $\bot$: it chooses that role's attempted moves arbitrarily, forever.

Transitions are exactly the instrumented rules of Section 3.4. The *value* $\mathrm{val}_{\mathcal{E}}(G, W_0)$ is the supremum over orchestrator strategies of the infimum over adversary strategies of the probability of reaching a goal-satisfying configuration.

Because modes modulate ε and modes change by history, the stochastic process seen by any fixed strategy pair is Markov-modulated, not i.i.d.: after a first deviation the process runs at the dr rate, which is the formal content of "deviations cluster." With no $\perp$-reachable states the game degenerates to an MDP; with $\varepsilon \equiv 0$ and $M_0 \equiv \mathsf{ok}$ it degenerates to the base transition system.

Proposition 1 (Conservativity)**.** *If $M_0 \equiv \mathsf{ok}$, $T_0 = \emptyset$, no capability has $\mathsf{prov} = \mathsf{ext}$, and $\varepsilon \equiv 0$, then (i) every instrumented rule agrees with its base counterpart on the reachable fragment, (ii) $G \vdash \langle \mathbb{M};\, W;\, M_0;\, d_0;\, \emptyset \rangle$ iff $G \vdash (\mathbb{M}; W)$, and (iii) $\mathrm{val}_{\mathcal{E}}(G, W_0) \in \{0, 1\}$ with value 1 iff $\Gamma; (G, W_0) \vDash \Diamond \varphi_{goal}$.*

Proof obligation. (i) and (ii) by rule-by-rule inspection (the closure premise is vacuous at $\varepsilon \equiv 0$ only in the semantics; in the typing it must be discharged — see Remark 2); (iii) by determinacy of reachability games. Mechanizable alongside T1.

Remark 2 (Cost of the closure premise). WIP. Even at $\varepsilon \equiv 0$, T-Comm-M/T-Act-M demand typability under downgraded modes — the static discipline insists the protocol *could* survive a deviation even if the profile says none occurs. This is a deliberate strictness (profiles are estimates; the type system should not trust them), but it means conservativity holds for the judgment only up to the closure obligations. Alternative: make the closure premise conditional on a pack-level flag. Co-author call.

6 Metatheory

Notation: Tn numbers are for this paper; they do not collide with the base paper's theorems. Each statement carries its proof obligation and its intended home (Coq vs. paper proof). Nothing below is discharged yet.

Theorem 1 (T1: Instrumented subject reduction)**.** *If $G \vdash K$ and $K \xrightarrow{\Lambda} K'$ by any instrumented rule, then $(G, W, M, d, T) \rightsquigarrow_{\Lambda} (G', W', M', d', T')$ and $G' \vdash K'$. In particular a $\mathsf{dev}(\mathsf{p})$ step preserves the type at the downgraded mode.*

Proof obligation. Extends the base subject-reduction proof (coinduction on typing, cases on the reduction rule); the deviant cases are new but easy — I-Dev is a stutter matched by G-Dev, and the closure premise of T-Comm-M/T-Act-M is exactly what makes the downgraded configuration typable. Coq target, on top of the (to-be-extended) multi-branch mechanization. Session fidelity extends symmetrically.

Theorem 2 (T2: Quarantine non-interference)**.** *For every run of the instrumented semantics from a typable initial configuration, under* every *behaviour of every $\perp$-role: no $a \in \mathsf{Irr}$ fires in a state where $\mathrm{supp}(pre_a) \cap T \neq \emptyset$, and no $a \in \mathsf{Irr}$ is fired by a role below dr.*

Proof obligation. Invariant argument over the instrumented LTS: the I-Act irreversibility premise is preserved by every rule; the demonic quantification is handled by I-Quar (all $\perp$ moves refused). This is the paper's headline safety theorem — "authorization-before-irreversible" as a typed guarantee, covering prompt injection without modelling the attacker. Coq target; expected to be the easiest of the majors.

Theorem 3 (T3: Mode monotonicity)**.** *(i) Along any run, $M(\mathsf{p})$ decreases except across I-Att steps naming p. (ii) Typability is closed under downgrade above $\perp$: if $G \vdash \langle \mathbb{M};\, W;\, M;\, d;\, T \rangle$ and $M' \sqsubseteq M$ pointwise with $M'(\mathsf{p}) \neq \perp$ wherever used by a non-T-Quar rule, then $G \vdash \langle \mathbb{M};\, W;\, M';\, d;\, T \rangle$. (iii) The graded value is monotone: $M' \sqsubseteq M$ implies* $\mathrm{val}_{\mathcal{E}} \leq$ *at M dominates* $\mathrm{val}_{\mathcal{E}}$ *at M'.*

Proof obligation. (i) by rule inspection; (ii) by coinduction, using that every premise is antitone in M; (iii) by strategy transfer in the game. (i)–(ii) Coq; (iii) paper. This is the quantitative sibling of the base capability-monotonicity theorem, and the continuity argument for reviewers.

Theorem 4 (T4: Graded soundness)**.** *If $\Gamma; \mathcal{E} \vdash \{S_{\mathsf{p}}\} : G \triangleright \Diamond_{\geq p}\varphi_{goal}$ then in $\mathcal{G}(\mathcal{P}, \mathcal{E})$ there is an orchestrator strategy achieving, against every adversary, goal probability at least p; moreover every run of that strategy satisfies the T2 safety invariant.*

Proof obligation. Soundness of the bound reduces to correctness of the value computation (T6) plus T1/T2; the interesting content is that conformance (qualitative) and value (quantitative) compose — the typed strategy space is nonempty and the value is computed over exactly the typed moves. Paper proof.

Theorem 5 (T5: Bounded degradation, and its converse)**.** *Let G satisfy WF-Loop and let every cycle contain a sanitizing $\checkmark\varphi$ or refresh for each role it exercises (the per-role refinement). Then there is a bound $p^* > 0$, independent of run length, with $\mathrm{val}_{\mathcal{E}}(G, W_0) \geq p^*$ whenever the loop-free skeleton achieves the goal with positive probability. Conversely, if some cycle of G exercises a role with neither refresh nor validation, and $\varepsilon(m, \cdot)$ is strictly increasing and bounded away from 0 at the cap D, then for the family $G^{(k)}$ of k-th unrollings, $\mathrm{val}_{\mathcal{E}}(G^{(k)}, W_0) \to 0$ as $k \to \infty$: no graded judgment certifies a nontrivial bound uniformly in horizon.*

Proof obligation. Positive direction: between consecutive checkpoints the grade is bounded by the cycle length, so per-cycle success probability is bounded below by a constant; a renewal argument (checkpoints as regeneration points of the modulated chain) gives horizon-independence. Negative direction: the grade saturates at D, per-step survival is at most $1 - \varepsilon(m, D) < 1$, and failures accumulate geometrically. Paper proof; the renewal argument is the technically substantive step. This is the theorem that *explains* long-horizon agent decay and prescribes the structural fix.

Theorem 6 (T6: Decidability and complexity)**.** *For finite* Pred*, finite-range Nd abstraction (the base checker's widened fragment), and a regular G: the state space of $\mathcal{G}(\mathcal{P}, \mathcal{E})$ has size $O(2^{|\mathsf{Pred}|} \cdot |\mathbb{M}_{\mathsf{c}}|^n \cdot (D+2)^n \cdot |G|)$ for n roles; the value of the simple stochastic game is computable, membership in $\mathrm{NP} \cap \mathrm{coNP}$ for the decision problem $\mathrm{val} \geq p$; and with no* `ext` *capability and no $\perp$-reachable state the game is an MDP whose value is computable in time polynomial in the (product) state space.*

Proof obligation. State-space bound by construction; SSG results are classical (Condon; Shapley) and imported, not re-proved; the MDP collapse by inspection of the player partition. Paper proof. This finally places a complexity statement in the discipline; the empirical scaling study must show real skills sit far from the $2^{|\mathsf{Pred}|}$ cliff.

Theorem 7 (T7: Undecidability boundary, inherited)**.** *Exact graded achievability (no widening, unbounded numerics) is undecidable; under dynamic topology it remains undecidable. The deviation layer does not restore decidability: the base reductions embed via Proposition 1.*

Proof obligation. Immediate from the base results once those proofs are written out fully (a known repair obligation in the base paper); the embedding is Proposition 1. Paper proof.

Mechanization plan. Coq: T1, T2, T3(i–ii), Proposition 1(i–ii), on top of the extended multi-branch base mechanization; all axiom-free, CI-pinned, with the same `Print Assumptions` harness discipline as the base development. Paper: T3(iii), T4, T5, T6, T7, importing classical SSG/MDP results by citation. We will not claim mechanized probability, and the paper will say so in one sentence rather than blur the line.

7 Falsifiable Predictions and Evaluation Plan

Each mechanism makes a prediction that measured traces can refute. We pre-register them here; the evaluation runs real agents (strong and weak models) against monitors compiled from corpus and real-skill protocols, logging deviation events per class of Section 3.1, with $n \geq 30$ runs per cell and bootstrap confidence intervals.

Prediction 1 (Modes: clustering). Deviation events cluster: a trace's post-first-deviation rate exceeds its pre-first-deviation rate. Test: fit i.i.d. vs. Markov-modulated deviation models to the trace corpus; likelihood-ratio test. Refutation of modulation refutes the mode structure (and simplifies the system — we would report that).

Prediction 2 (Grades: depth dependence and reset). The hazard of a first deviation increases with steps since session start, and drops after a refresh action (context compaction / spec re-read). Test: survival analysis on the same traces, refresh as a covariate.

Prediction 3 (Demonic bottom: injection is not noise). Under a prompt-injection arm, (a) injected runs violate any ε-bound calibrated on clean runs (the probabilistic layer alone is falsified exactly where it should be), and (b) across all injected runs, gated per T2, no Irr capability fires on tainted support — zero occurrences, not a rate. Test: adversarial arm with seeded injections in `ext` payloads.

The calibration product is the compliance profile $\mathcal{E}$ per model and task family, published with the artifact; a model-strength sweep bounds the "our enforcement matters" claim (strong models approach $\varepsilon \to 0$ on shallow tasks — we will say so before a reviewer does). The predictive-validity experiment closes the loop: predicted $\Diamond_{\geq p}$ bounds from calibrated profiles against observed success rates, as a calibration plot. Either outcome is reportable; a divergence localized at long horizons would itself confirm Prediction 2's mechanism.

8 Related Work

WIP: Compressed here; the submission version engages each line in prose. The bibliography keys below are placeholders to be completed — the base paper's known citation-key mis-resolution (`synthetic2023`/`synthetic2026`) must be fixed before any of this text is reused.

Probabilistic session types put probability on internal choices of compliant processes (binary probabilistic sessions [2, 1]; probabilistic refinement and resource-aware session types [3, 4]; interval-probability MPST [5]). Our probability attaches to *conformance*, not choice; the type syntax stays qualitative. *Failure-tolerant MPST* (crash-stop [6], affine/exceptional [7]) model participants that stop; deviation generalizes crash (δ_{sil} is one class among five). *Monitoring*: the decentralized-monitor theorem [8] restores safety at the boundary; our delta is goal achievability under deviation, plus taint and irreversibility. *Information-flow session types*: the mode/taint layer descends directly from access- and information-flow control for sessions [9], redirected from confidentiality to trust in the participant itself. *Synthetic MPST* [10, 11]: we adopt the projection-free line; novelty weight sits on the world/goal/deviation layers, and the precision relationship (plain merge vs. full merge) is stated, not claimed away. *LTL compliance monitoring of LLM agents* [12, 13]: trace-level temporal properties with learned propositional abstraction and sampling-based predictive intervention; complementary on three axes — multiparty vs. single-agent, deterministic pre-emission gating vs. predicted-violation intervention, and a deterministic trusted core vs. LLM labelers in the checking path. Their finding that violations concentrate in ordering/sequencing is independent evidence for the structural layer.

9 Honest Scoping

(1) The mode lattice is three-point and the grade cap small by design; the state-space factor $|\mathbb{M}_{\mathsf{c}}|^n(D+2)^n$ is affordable for skill-sized protocols and must be shown affordable empirically, not asserted. (2) The κ kernel and ε are estimates; all quantitative claims are conditional on the profile, which is why safety (T2) is deliberately profile-independent. (3) δ_{pay} detection is only as strong as

the $\checkmark\varphi$ nodes the author writes; an unvalidated latent corruption that never crosses an Irr guard is out of scope (it is also, by definition, never irreversible). (4) The quarantine residual (Remark 1) is an open design point. (5) Mechanization covers the qualitative layer only. (6) Everything here presumes the gate architecture: an agent with un-gated side channels to the world voids T2's premise — the theorem is about what the declared capability surface can do, which is exactly the base discipline's trust boundary.

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